## Introduction

Meta-regression extends the random-effects meta-analysis by allowing study-level covariates — moderators such as publication year, mean participant age, average dose, or risk-of-bias score — to enter the model as predictors of the underlying effect. The fitted slope quantifies how much the true study effect changes per unit of moderator, and a residual between-study variance $\tau^2$ describes whatever heterogeneity the moderator did not explain. Meta-regression is the appropriate tool whenever the candidate moderator is continuous, and is generally more powerful than dichotomising into subgroups, which discards information and can produce a categorisation-dependent answer. It is now standard practice for any meta-analysis with $k \geq 10$ studies and a pre-specified hypothesis about effect modification.

## Prerequisites

A working understanding of random-effects meta-analysis (notably the role of $\tau^2$), ordinary linear regression, and the difference between fixed-effect, random-effects, and mixed-effects models in the meta-analytic context.

## Theory

The mixed-effects meta-regression model is

$$y_i = \beta_0 + \beta_1 x_i + u_i + \varepsilon_i, \qquad u_i \sim N(0, \tau^2), \quad \varepsilon_i \sim N(0, s_i^2),$$

where $u_i$ is between-study residual heterogeneity and $\varepsilon_i$ is within-study sampling error with known variance $s_i^2$. REML estimates $\tau^2$ jointly with the regression coefficients; Wald or likelihood-ratio tests assess moderator effects. The pseudo-$R^2 = 1 - \tau^2_{\text{residual}}/\tau^2_{\text{null}}$ summarises the proportion of between-study variance explained by the moderator and is the analogue of the regression $R^2$.

## Assumptions

The moderator is pre-specified rather than data-mined; the functional form (linear unless explicitly modelled otherwise) is correct; the residual heterogeneity is normal; and there are sufficient studies — a frequently-cited rule of thumb is at least ten studies per moderator coefficient — to support reliable estimation.

## R Implementation

```r
library(metafor)

set.seed(2026)
k <- 15
year <- sample(1995:2023, k, replace = TRUE)
mean_age <- runif(k, 40, 70)

yi <- 0.5 - 0.02 * (year - 1995) + rnorm(k, 0, 0.15)
vi <- 1 / sample(50:250, k, replace = TRUE)

df <- data.frame(yi, vi, year, mean_age)

res1 <- rma(yi, vi, mods = ~ year, data = df, method = "REML")
summary(res1)

res2 <- rma(yi, vi, mods = ~ year + mean_age, data = df, method = "REML")
summary(res2)

cat("R^2:", round(res1$R2, 3), "\n")
```

## Output & Results

`rma()` with a `mods` formula returns moderator coefficients with confidence intervals and $p$-values, the residual heterogeneity statistics ($\tau^2$, $I^2$, residual $Q_E$), an omnibus moderator test ($Q_M$), and the pseudo-$R^2$ summarising explained heterogeneity. Reporting all four elements — slope, residual $\tau^2$, $Q_M$, and $R^2$ — is standard practice.

## Interpretation

A reporting sentence: "Year of publication significantly moderated the pooled effect (slope $-0.017$, 95 % CI $-0.029$ to $-0.005$, $p = 0.01$): each additional year reduced the standardised mean difference by 0.017 units. Meta-regression explained 52 % of the between-study variance ($R^2$); residual heterogeneity remained moderate ($I^2_{\text{residual}} = 21\%$). The temporal trend was pre-specified in the PROSPERO protocol." Always report both the slope and the residual heterogeneity to convey what the moderator did and did not explain.

## Practical Tips

- Restrict to roughly one moderator per ten studies; meta-regression overfits aggressively because between-study sample sizes are tiny by regression standards.
- Pre-specify all moderators in the PROSPERO protocol; post-hoc moderators are at best exploratory and should be flagged as such in reporting.
- Centre continuous moderators (e.g., year minus mean year) so the intercept has a meaningful interpretation as the effect at the mean moderator value.
- For small $k$, permutation tests via `permutest()` provide more conservative inference than the Wald-based default and are widely recommended.
- Beware ecological fallacy: a study-level moderator effect does not imply the same effect at the individual-participant level; that requires IPD meta-analysis.
- Plot bubble plots (`metafor::regplot()`) showing study effects, moderator value, and weighted regression line; they make the moderator relationship visible and reveal influential outliers.

## R Packages Used

`metafor::rma()` with `mods = ~ ...` for the canonical mixed-effects meta-regression; `metafor::regplot()` for bubble plots; `metafor::permutest()` for permutation-based inference under small $k$; `meta::metareg()` for the alternative `meta`-package interface; `robumeta` for robust-variance meta-regression with dependent effect sizes.

## For Reviewers

**What to look for in a paper using this method.**

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

## See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis